POSTS AND TELECOMMUNICATIONS INSTITUTE OF TECHNOLOGY

Faculty / Department of Information Technology

SEMESTER EXAMINATION

INTELLIGENT SYSTEM DEVELOPMENT

ORAL EXAMINATION

Question Bank

50 Questions

Learning Materials:
Students have completed the consolidated learning material `tieuluanFinal`, covering:

- Machine Learning and Deep Learning;
- Convolutional Neural Networks (CNN);
- Recurrent Neural Networks (RNN) and LSTM;
- Knowledge Graphs and Neo4j;
- Retrieval-Augmented Generation (RAG);
- Intelligent Agents and Multi-Agent Systems.

Practical Development Experience:
Students have developed and deployed Deep Learning applications for:

- Diabetes prediction;
- Stock price prediction;
- House price prediction.

Students are also developing the final project:

Multi-Agent Development for Diabetes
An intelligent multi-agent system integrating Deep Learning, Knowledge Graphs, RAG, and agent-based assistance.

Examination Objectives

The oral examination evaluates students' abilities to:

1. Understand fundamental concepts and algorithms in Intelligent System Development.
2. Explain the mathematical and computational foundations of Machine Learning and Deep Learning.
3. Design and implement CNN, RNN, and LSTM models.
4. Represent domain knowledge using Knowledge Graphs and Neo4j.
5. Construct RAG-based intelligent applications.
6. Understand agent architectures, cooperation, and multi-agent coordination.
7. Integrate AI models, knowledge bases, and agents into practical intelligent applications.
8. Explain, demonstrate, and defend their own project implementations.

Examination Method

1. Students prepare their answers using their laptops.
2. Students may present their learning materials, diagrams, source code, and experimental results.
3. The examiner selects questions from the question bank.
4. Students answer orally at the examination table.
5. The examiner may ask additional questions concerning algorithms, source code, architecture, and experimental results.
6. Students must demonstrate their actual project implementation when requested.

1. Machine Learning and Deep Learning

Focus: Fundamental principles, mathematical foundations, algorithms, model training, and practical prediction applications.

Question 1. Explain the concept of Machine Learning. Distinguish supervised learning, unsupervised learning, and reinforcement learning. Give a practical example of each approach.

Question 2. Describe the complete workflow for developing a Machine Learning prediction model. Explain data collection, preprocessing, feature engineering, model training, validation, and testing.

Question 3. What is the difference between classification and regression? Explain their differences using diabetes prediction, stock prediction, and house price prediction.

Question 4. Explain the architecture of an Artificial Neural Network. What are the roles of input layers, hidden layers, output layers, weights, and biases?

Question 5. Explain forward propagation and backpropagation in a neural network. How are gradients calculated and used to update the network parameters?

Question 6. What is a loss function? Explain the differences between Mean Squared Error (MSE), Binary Cross-Entropy, and Categorical Cross-Entropy. Give an appropriate application for each.

Question 7. Explain Gradient Descent and its variants: Batch Gradient Descent, Stochastic Gradient Descent, and Mini-Batch Gradient Descent. What is the role of the learning rate?

Question 8. What are overfitting and underfitting? Explain the use of training, validation, and testing datasets. Discuss regularization, dropout, and early stopping.

Question 9. How do you evaluate a classification model? Explain accuracy, precision, recall, F1-score, and confusion matrix. Why can accuracy be misleading for imbalanced diabetes datasets?

Question 10. Compare classification and regression models in practical applications. Explain the main challenges of deploying a trained Deep Learning model as a real-world web application or API.

2. CNN, RNN and LSTM

Focus: Neural network architectures, data representation, feature extraction, and sequential learning.

Question 11. Explain the architecture of a Convolutional Neural Network (CNN). What are convolutional layers, filters, feature maps, activation functions, and pooling layers?

Question 12. Explain how a convolution operation works. Describe the roles of kernel size, stride, padding, and the number of filters.

Question 13. Why are CNNs effective for image classification? Explain how a CNN learns features at different levels, from basic patterns to complex structures.

Question 14. Describe the process of developing a CNN-based image classification application. Explain image preprocessing, training, validation, inference, and deployment.

Question 15. What is a Recurrent Neural Network (RNN)? Explain its hidden state and how information is propagated through a sequence.

Question 16. What are the vanishing gradient and exploding gradient problems in RNNs? Why do these problems affect long-term dependency learning?

Question 17. Explain the architecture of Long Short-Term Memory (LSTM). Describe the functions of the forget gate, input gate, output gate, cell state, and hidden state.

Question 18. Compare RNN, LSTM, and CNN. Discuss their strengths, limitations, and suitable applications. Explain why RNN or LSTM may be appropriate for stock time-series prediction.

3. Knowledge Graphs, Neo4j and RAG

Focus: Knowledge representation, graph databases, Cypher queries, information retrieval, and knowledge-grounded generation.

Question 19. What is a Knowledge Graph? Explain entities, relationships, properties, and attributes. Give an example from the diabetes domain.

Question 20. Explain the graph representation of knowledge using subject-predicate-object triples. Construct at least five triples for a simple diabetes Knowledge Graph.

Question 21. What is Neo4j? Explain the property graph model. How are nodes, relationships, and properties stored and represented?

Question 22. Explain the basic Cypher query language. Write Cypher queries to create a Patient, a Disease, and a relationship between them. Then retrieve the corresponding information.

Question 23. Design a simple Knowledge Graph for diabetes. Identify appropriate entities and relationships among patients, symptoms, risk factors, measurements, diseases,

and recommendations.

Question 24. Explain how Neo4j can support knowledge retrieval and reasoning. Give an example of traversing multiple relationships to retrieve relevant information.

Question 25. What is Retrieval-Augmented Generation (RAG)? Explain why a RAG system combines information retrieval with a Large Language Model (LLM).

Question 26. Describe the complete architecture of a RAG system. Explain document ingestion, text chunking, embedding generation, vector storage, retrieval, and response generation.

Question 27. Compare traditional keyword retrieval, vector similarity search, and Knowledge Graph retrieval. What are their advantages and limitations?

Question 28. How can Neo4j Knowledge Graphs and RAG be integrated into a diabetes intelligent assistant? Explain how the system retrieves relevant knowledge and generates grounded responses.

4. Intelligent Agents and Multi-Agent Systems

Focus: Agent architecture, autonomous decision-making, communication, cooperation, task decomposition, and system coordination.

Question 29. What is an Intelligent Agent? Explain the concepts of environment, perception, state, action, goal, and decision-making.

Question 30. Describe the general architecture of an Intelligent Agent. What are the roles of the agent's input, internal processing, knowledge, decision mechanism, and output?

Question 31. Distinguish a single-agent system from a Multi-Agent System (MAS). What problems can be solved more effectively through multiple cooperating agents?

Question 32. Explain the architecture of a Multi-Agent System. Describe agent roles, communication, coordination, cooperation, and task allocation.

Question 33. What is the difference between centralized, decentralized, and hierarchical multi-agent architectures? Discuss the advantages and disadvantages of each architecture.

Question 34. Explain how agents can communicate and exchange information. Describe message-based communication, shared knowledge, and task delegation.

Question 35. What is Reinforcement Learning? Explain the concepts of state, action, reward, policy, and environment. How can an agent learn through interaction?

Question 36. Describe the Markov Decision Process (MDP). Explain the state space, action space, transition probability, reward function, and policy.

Question 37. What are the differences between an AI agent and a traditional software component? Discuss autonomy, goal-directed behavior, reasoning, tool usage, and interaction with other agents.

Question 38. Explain the main challenges in developing a Multi-Agent System. Discuss coordination, communication, conflicting decisions, reliability, security, and performance evaluation.

5. Practical Applications and Diabetes Multi-Agent Project

Focus: Application development, system integration, deployment, project architecture, and practical demonstration.

The following questions are directly related to students' practical experience and final project.

A. Deep Learning Applications

Question 39. Explain your diabetes prediction application. What dataset did you use? What are the input features, target variable, Deep Learning architecture, and prediction output?

Question 40. Explain how you developed and deployed your diabetes prediction application. Describe the interaction between the user interface, backend, trained model, and prediction service.

Question 41. Compare your diabetes prediction application with your stock price prediction and house price prediction applications. Explain the differences in input data, prediction objectives, model outputs, and evaluation metrics.

Question 42. Demonstrate how a new input sample is processed by one of your deployed Deep Learning applications. Trace the process from user input to prediction output.

B. Diabetes Multi-Agent System Architecture

Question 43. Present the overall architecture of your Multi-Agent Development for Diabetes project. Explain the main components and their relationships. Why did you choose a Multi-Agent System instead of a single-agent architecture?

Question 44. Identify the agents in your diabetes Multi-Agent System. For each agent,

explain its responsibility, input, output, and interaction with other agents.

Question 45. Explain how your Deep Learning diabetes prediction model is integrated into the Multi-Agent System. Which agent invokes the prediction model? How is the prediction result communicated to other agents?

Question 46. Explain how the Knowledge Graph and Neo4j are used in your diabetes project. What entities and relationships have you implemented? Demonstrate an example of a Cypher query used by your system.

Question 47. Explain how RAG is integrated into your diabetes assistant. How does the system retrieve relevant information and generate a response based on its knowledge base? How do you reduce unsupported or hallucinated answers?

Question 48. Describe the complete execution flow of your Multi-Agent System when a user asks a question about diabetes. Explain the roles of the coordinator agent, prediction agent, knowledge retrieval agent, and response agent, if these components are present in your implementation.

Question 49. Demonstrate your completed Multi-Agent System on your laptop. Choose one realistic user scenario and explain the processing steps, agent interactions, retrieved knowledge, and final response.

Question 50. How do you evaluate your Multi-Agent System? Discuss the evaluation of the Deep Learning model, knowledge retrieval quality, response accuracy, agent coordination, response time, and overall system reliability.

Question 51. Discuss the limitations and potential risks of your diabetes Multi-Agent System. How would you improve the architecture, agent coordination, knowledge quality, security, and deployment in a future version?

Guidelines for Students

Students should prepare the following materials before the oral examination:

1. **Learning material:** The completed `tieuluanFinal`, including all theoretical and technical chapters.
2. **Source code:** The source code of their Deep Learning applications and Multi-Agent System.
3. **Datasets:** Information about the datasets, features, preprocessing, and experimental setup.
4. **Model results:** Training results, evaluation metrics, prediction examples, and model limitations.
5. **Knowledge Graph:** Neo4j database structure, graph schema, sample nodes, relationships, and Cypher queries.
6. **RAG implementation:** Document sources, chunking strategy, embedding model, retrieval process, and generated responses.
7. **Multi-Agent System:** Architecture diagram, agent definitions, communication mechanisms, execution flow, and implementation.
8. **Project demonstration:** A working application demonstrating the actual functions implemented by the student.

Oral Examination Requirements

1. Students must understand the questions and explain the answers in their own words.
2. Students should use mathematical expressions, diagrams, examples, and source code when appropriate.
3. Students must be able to explain the design decisions and implementation of their own applications.
4. Students should distinguish between implemented functions and proposed future functions.
5. Students must not claim that an agent, model, or integration has been implemented unless it can be demonstrated or verified.
6. For diabetes-related applications, students must explain the limitations of predictive models and the importance of reliable medical knowledge and professional healthcare supervision.

Suggested Assessment Criteria

Assessment Component	Weight
Theoretical understanding of ML and DL	15%
Understanding CNN, RNN, and LSTM	10%
Knowledge Graph, Neo4j, and RAG	15%
Intelligent Agents and MAS	15%
Project architecture and implementation	20%
Practical demonstration and technical defense	15%
Critical thinking and problem-solving	10%
Total	**100%**

END OF ORAL EXAMINATION QUESTION BANK

Intelligent System Development